Hwk 2 : solutions

1. $f = -\frac{2m}{\hbar^2}\int_0^\infty V(r)\frac{\sin qr}{q}\, r\, dr$

(i) Square well $\frac{d\sigma}{d\Omega} = 4a^2\left(\frac{-mV_0a^2}{\hbar^2}\right)^2\frac{\left[\sin(qa) - qa\cos(qa)\right]^2}{(qa)^6}$

Let $q = 2k\sin\left(\frac{\theta}{2}\right)$ then

$\sigma = \frac{2\pi}{k^2}\left(\frac{mV_0a^2}{\hbar^2}\right)^2\left[1 - \frac{1}{(2ka)^2} + \frac{\sin(4ka)}{(2ka)^3} - \frac{\sin^2(2ka)}{(2ka)^4}\right]$

This is not required, only asked for high energy limit which is $\sigma \sim \frac{2\pi}{k^2}\left[\frac{mV_0a^2}{\hbar^2}\right]^2$, $ka \gg 1$

(ii) Gaussian — in my notes!

2. (a) Compute $ka = \sqrt{2mE}\, a/\hbar = \sqrt{2mc^2E}\, a/\hbar c$

$mc^2 = 938\,\text{MeV}$, $E = 1\,\text{MeV}$, $a = 10^{-13}\,\text{cm} = 1\,\text{fm}$, $\hbar c = 1.97\times10^{2}\,\text{MeV}\cdot\text{fm}$

$\Rightarrow ka = .220 < 1$. For small ka not near a resonance

$\delta_\ell \sim (ka)^{2\ell+1}$, so $f(\theta) \sim \frac{1}{2ik}\left(2i\delta_0 + 2i\delta_1 P_1\cos\theta + 2i\delta_2 P_2 - \cdots\right)$

then $\frac{d\sigma}{d\Omega} \sim \delta_0^2,\ \delta_0\delta_1,\ \delta_1^2,\ \delta_0\delta_2$

$\sim (ka)^2,\ (ka)^4,\ (ka)^6,\ (ka)^6$

$\Rightarrow 1 : (ka)^2 : (ka)^4 : (ka)^4 \Rightarrow 1 : 0.05 : .001$

So to 1% accuracy, best we can do is determine δ_0, δ_1.

You may also do this as I suggested by explicitly computing error propagation. I attach that too — next page!

(b) $\sigma = 4\pi/k^2\left(\sin^2\delta_0 + 3\sin^2\delta_1\right)$ $\left|\frac{d\sigma}{d\Omega} = k^{-2}\left|\cos\delta_0\sin\delta_0 + i\sin^2\delta_0 + 3\left(\cos\delta_1\sin\delta_1 + i\sin^2\delta_1\right)\times\right.\right.$

$\left.\left.\text{at } \theta = \frac{\pi}{2}\ \cos\theta = 0 \Rightarrow \delta_0 \simeq 0.20,\ \text{From } \sigma,\ \sin^2\delta_1 = 2.14\times10^{-4}\cos\theta\right|$

$\Rightarrow \delta_1 \sim 0.01$

2 — Other solution: Compute $a\left[\frac{d\sigma}{d\Omega}\right]$ to get error in δ_i

$$\frac{d\sigma}{d\Omega} \sim \delta_0^2 + \delta_1^2 + \delta_2^2 + 2\delta_1\delta_2 + 2\delta_0\delta_1 + 2\delta_0\delta_2$$

$$d\left(\frac{d\sigma}{d\Omega}\right) \sim \sum_{i=1}^{3} \frac{\partial\left(d\sigma/d\Omega\right)}{\partial\delta_i} d\delta_i = \left(\sum_i \delta_i\right)\left(\sum_j d\delta_j\right)$$

Fractional error in $\frac{d\sigma}{d\Omega}$ is $\frac{d\left(d\sigma/d\Omega\right)}{d\sigma/d\Omega} \sim \frac{\sum_i d\delta_i}{\sum_j \delta_j} \sim \frac{\sum_i d\delta_i}{\delta_0}$

Thus, if $\frac{d\left(d\sigma/d\Omega\right)}{d\sigma/d\Omega} \equiv \epsilon (\sim 0.01)$ then $\epsilon\delta_0 \sim \sum_i d\delta_i$

Then $\frac{d\delta_i}{\delta_i}$ the relative error in $\frac{\epsilon\delta_2}{\delta_i} \sim \epsilon (ka)^{-2\ell}$

Then error in $\delta_0 \sim \epsilon$ (1%)

$$\delta_1 \sim \frac{\epsilon}{.05} \quad (20\%)$$

$$\delta_2 \sim \frac{\epsilon}{.002} \quad (\sim 500\%!)$$

Only δ_0, δ_1 can be determined

(b) $\sigma = \frac{4\pi}{k^2}\left(\sin^2\delta_0 + 3\sin^2\delta_1\right)$

$$\frac{d\sigma}{d\Omega} = k^{-2}\left|\cos(\delta_0)\sin(\delta_0) + i\sin^2\delta_0 + 3\left(\cos\delta_1\sin\delta_1 + i\sin^2\delta_1\right)\cos\theta\right|^2$$

at $\theta = \pi/2$ $\cos\theta = 0$ so $.823 \times 10^{-26}\,cm^2 = k^2\sin^2\delta_0$

$$\Rightarrow \delta_0 \approx 0.20$$

From σ, $\sin^2\delta_1 = \frac{1}{3}\left(\frac{k^2\sigma}{4\pi} - \sin^2\delta_0\right) = 2.14 \times 10^{-4} \Rightarrow \delta_1 \sim 0.01$

3. i) $\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$

$\delta_l = \sin^{-1}\left[\frac{(iak)^l}{\sqrt{(2l+1)}\, l!} \right]$ $\quad \sin^2 \delta_l = \frac{(-a^2k^2)^l}{(2l+1)\, l!}$

Then $\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} \frac{(-a^2k^2)^l}{l!} = \frac{4\pi}{k^2} \exp\{-a^2k^2\}$

$= \frac{4\pi\hbar^2}{2mE} \exp\left\{ -\frac{2mEa^2}{\hbar^2} \right\}$

b) ka small

4) $\rho = \frac{1}{z} e^{-ax - bx^2}$ Let $\mu_0 = 1 = \int_{-\infty}^{\infty} dx \rho(x) = \frac{1}{z}\sqrt{\frac{b}{a}}\, e^{a^2/4b}$

$\mu_1 = \int_{-a}^{a} dx\, x\rho = \frac{\sqrt{\pi}}{2b^{3/2} z} e^{a^2/4b}$ $\quad \mu_2 = \frac{\sqrt{\pi}(2b+a^2)}{4b^{5/2} z}$

$\frac{\mu_2}{\mu_1} = \frac{2b+a^2}{2ab}$ $\quad \frac{\mu_1}{\mu_0} = \frac{a}{2b}$. Define $\sigma^2 = \mu_2 - \mu_1^2$ then

$a = \mu_1/\sigma^2$ $\quad b = \frac{1}{2\sigma^2}$ This, along with $z = \sqrt{\frac{\pi}{b}} e^{a^2/4b}$ yields

$\rho = \frac{1}{\sqrt{2\pi}\sigma} \exp -\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$ The normal distribution!